

HydroSentinel Hardware Schematics & Pinouts

Target Hardware: ESP32-S3 | JSN-SR04T | DHT22 | SX1278 Ra-02 LoRa | SSD1306 OLED

Document Type: Clean Connection Specifications

1. Edge Sensing Node (Transmitter) Architecture

Description: Autonomous edge post. A 5V solar panel charges a lithium battery via a TP4056 module, boosted to 5V/3.3V by an MT3608 converter. The ESP32-S3 adjusts water distance measurements using DHT22 temperature data. The JSN-SR04T ultrasonic sensor operates via a bi-directional logic converter, and broadcasts alert packets over 433 MHz LoRa.

POWER SUBSYSTEM Solar 5V + TP4056 + MT3608 Boost	ENVIRONMENT SENSOR DHT22 (3.3V Logic)	WATER LEVEL SENSOR JSN-SR04T + Logic Conv.	LORA TRANSCEIVER SX1278 (433 MHz SPI)
↓ Output: 5V & 3.3V Rails	↓ Data: GPIO 4	↓ Trig: GPIO 5 Echo: GPIO 18	↓ SPI: GPIO 15, 23, 19, 18
ESP32-S3 MICROCONTROLLER BOARD System Core & Edge Processing Unit			

Component	Pin	ESP32-S3 Connection	Description
JSN-SR04T	VCC / GND	5V Rail / Common GND	Sensor logic power and ground reference
JSN-SR04T	TRIG	GPIO 5 (via Logic Conv.)	Trigger pulse line (3.3V to 5V step-up)
JSN-SR04T	ECHO	GPIO 18 (via Logic Conv.)	Echo pulse return (5V to 3.3V step-down)
DHT22	DATA	GPIO 4	Climate data pin with 10k pull-up resistor
SX1278 LoRa	NSS / CS	GPIO 15	SPI Chip Select line
SX1278 LoRa	MOSI / MISO	GPIO 23 / GPIO 19	SPI Master Output and Slave Input lines
SX1278 LoRa	SCK / RST	GPIO 18 / GPIO 14	SPI Clock line and Hardware Reset pin
SX1278 LoRa	DIO0	GPIO 2	Hardware interrupt for incoming/outgoing packets

2. Base Station Gateway (Receiver) Architecture

Description: Powered via 5V DC stepped down to 3.3V with an AMS1117 regulator. Listens for incoming 433 MHz LoRa packets, displaying real-time metrics on a SSD1306 OLED screen. Upon receiving a hazard broadcast, it drives a high-decibel 12V siren via a 2N2222 transistor relay and triggers visual LEDs.

OLED DISPLAY SSD1306 (I2C)	ALARM RELAY 2N2222 Transistor	STATUS LEDs Green / Red	LORA RECEIVER SX1278 (433 MHz)
↓ I2C: GPIO 21 / 22	↓ Trigger: GPIO 12	↓ GPIO 16 / 17	↓ SPI: GPIO 15, 23, 19, 18
ESP32 / ESP8266 BASE STATION GATEWAY Alert Processing Unit			

Component	Pin	ESP32 Connection	Description
SX1278 LoRa	VCC / GND	3.3V Rail / Common GND	3.3V power input & ground
SX1278 LoRa	SPI Pins	GPIO 15, 23, 19, 18	CS, MOSI, MISO, and SCK lines
SSD1306 OLED	SDA / SCL	GPIO 21 / GPIO 22	I2C Serial Data and Clock lines
Siren Relay	Base Control	GPIO 12	Drives 2N2222 NPN transistor with 1k resistor
Status LEDs	Green / Red	GPIO 16 / GPIO 17	System OK (Green) & Hazard Alert (Red) via 220Ω resistors